MA7442 Assessment 1 Guide

Submission deadlines

- Task 1 - Friday 13th February 2026 at 12pm (midday)
- Task 2 - Friday 20th March 2026 at 12pm (midday)

Coursework weighting

- 80%

Scope

The assessment requires you to either: i) choose from one of five pre-selected data sets covering different disciplines OR ii) obtain permission from the module convenor to use your own data set of interest. The document containing details of the data sets and the location of these is provided separately.

If you wish to use your own data set. Please discuss this with a module convenor at an early stage of the project. Any data set which is used must be: i) fairly large (>10,000 records); ii) freely available to use for academic purposes; iii) contain multiple variables allowing tasks to be undertaken.

You will undertake two tasks which will be marked separately. You will receive and use feedback and guidance on task 1 to support task 2. However, you will be able to start work on task 2 prior to receiving this feedback.

Assessments will be marked according to the rubrics provided in the Excel file available on Blackboard. This includes criteria for both Task 1 and 2 (written submissions and interview).

The total assessment is worth 80% of the marks for the module and includes the following tasks:

Task 1. Initial exploratory data analysis and research questions (20%)

- Provide a written description and exploratory data analysis of your data set and any key points of interest which should be considered.
- Use your evaluation to develop a set of research questions.
- Produce aims and objectives which focus on answering three research questions. At least two of these research questions should involve use of data modelling approaches. The other can include more general data science approaches.
- Relevant research questions will be based on real-world scenarios not method evaluation. For instance, questions should not ask if one type of modelling approach is better than another.

Task 2. Modelling project (60%):

- Reflect on feedback from task 1 and finalise research questions. If necessary, include additional data analyses.
- Answer your research questions using R or Python and your dataset.
- Identify the rationale for using specific analytical/modelling approaches to answer your research questions and the strengths and limitations of the approaches you use. Multiple model types should be implemented and evaluated where it makes sense.
- Provide a clear description of the methodological process which could be repeated by someone else and describe why you have used particular approaches.

- Critically analyse your results in relation to current research in the area, including researching associated literature and citing peer-reviewed publications. This should also include considerations of sustainability, ethical and EDI issues.
- Produce a well-formatted, clearly and concisely written report.
- Undertake a 5-minute interview in which you will be asked questions about your project. This will be organised during week 10 of the module.

A template for the reports is not provided. At the MSc level you will be expected to produce a suitable structure based on your experience, knowledge and provided assessment criteria.

Submission details

Format: Submissions should be completed as two well-formatted Quarto documents (one for each task) using R or Python programming languages. PDFs should be generated from the Quarto document. Both the PDF and the Quarto markdown files should be submitted together. Missing files, submitted PDF reports not generated from source Quarto files and/or submission of non-Quarto files will receive mark deductions.

The two files should be submitted on Blackboard using the appropriate link in the Assessment and Feedback folder. Quarto markdown should be used to format tables, include citations etc. References should be appropriately cited and an appropriate referencing format should be used (e.g. Harvard, IEEE). The generated PDF reports should be well produced (e.g. not include irrelevant boiler-plate text, un-necessary code or errors) with appropriate Markdown-based documentation surrounding the code sections.

Submitted files should be named: `MA7442_A1T1_[student_number]` (for task 1) and `MA7442_A1T2_[student_number]` (for task 2).

Late submissions: These will receive mark deductions in line with university regulations (https://le.ac.uk/policies/regulations/senate-regulations/senate-regulation-7), unless approved by the mitigating circumstances team.